INDY PRIETO

Ames, IA | iprieto@iastate.edu

SUMMARY

Highly motivated and *highly* adaptable. Loving nothing more than a good challenge, I routinely place myself on the cutting-edge to deliver real and competitive results. Using a mixture of multi-field software and cybersecurity engineering experience alongside a wide swath of general knowledge, I specialize in turning "dreams" into "done".

EDUCATION

Iowa State University, Ames, IA — Expected Graduation 2026

- *BS in Cybersecurity Engineering*
- *Awards:* Best Coder, Principal Engineering Scholarship, Maurice S. and Elinor J. Reid Endowed Scholarship in Electrical and Computer Engineering, Richard K Richards scholarship in Electrical and Computer Engineering, ISU scholarship for student achievement, 1st place ISU CDC 2

TECHNICAL SKILLS

- *Computer Languages: Python, Java, Rust, C, Assembly. Familiar with many others.*
- *Tools: VSCode, Docker, Git, Wireshark, Burp CE*
- *Skills: Software Engineering, Artificial Intelligence System Design, Vulnerability Analysis, Embedded Systems, Networking, Controls*

WORK EXPERIENCE

Iowa State University, Ames, IA — 2024 - 2025

Undergraduate Researcher (Summer 2025 - Present)

- Work developing a complex secure E2E communication protocol for the smart grid

Teaching Assistant for Cryptographic Applications (2024)

- Led/Co-Led a lab section, graded student work, and administrated online student communications

Argonne National Lab, Chicago, State — 2023 - 2024

Research associate (Summer 2024)

- Work regarding cybersecurity sandboxes, and embedding
- Co-authored research published at: https://doi.org/10.1109/IECON55916.2024.10905821

SULI Intern (Summer 2023)

- Work regarding symbolic AI and theoretical combination with LLMs alongside education on research skills
- Co-authored (First author) research published at: https://doi.org/10.1609/aaaiss.v2i1.27704

PROJECTS & LEADERSHIP EXPERIENCE

Space Anomaly Detection, *Senior Project* — 2024 - 2025

- Project to investigate the possible use of edge-based AI to detect space anomalies on a satellite
- Overall designer, creating a system that used a cutting-edge LLM, a custom DSPy backend (spun off), and a natural-language searchable database, tested on simulated communications via NS-3.

MIPS CPU, *Class Project*, Computer Organization and Assembly Level Programming — 2023

- Used VHDL to develop a pipeline-based CPU, with speculative execution, with a team
- Personally developed a solution to speed up arduous test-creation and a python-based VHDL source code generator to design an arbitrarily-sized enhanced adder system that has fundamentally higher performance

CyBot, *Class Project*, Embedded Systems I: Introduction — 2023

- Used C (and Godot) to make a functional user-controlled robot that can complete tasks.
- Used a game development engine, Godot, to rapidly prototype and deploy a complete control system that allowed game-controller manipulation of a small robot, with instant feedback regarding environment scans and contact sensors on an intuitive interface.